DESIGN AUDIT AGENT

Transforming Screenshots into Actionable Design Intelligence through Responsible AI

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1.Executive Summary

Modern software teams continuously review user interfaces, evaluate redesigns, and validate releases before deployment. These activities are often manual, subjective, and time-consuming.

Design Audit Agent was developed to demonstrate how AI-assisted agents can support design evaluation workflows while maintaining explainability, transparency, and human oversight.

The solution is implemented as a three-level architecture:

• Level 1 – Design Audit

• Level 2 – Design Comparison

• Level 3 – Autonomous Regression Agent

Together, these agents transform screenshots into actionable insights that help teams improve usability, evaluate design changes, and support release decisions.

2.Problem Statement

Product teams frequently face challenges such as:

• Determining whether a UI is usable

• Evaluating whether redesigns improve user experience

• Identifying regressions before release

• Prioritizing design improvements

Traditional design reviews rely heavily on manual inspection and subjective interpretation. As products scale, maintaining consistency and review quality becomes increasingly difficult.

The objective of this project is to explore how AI can augment design review workflows without replacing human judgment.

3.Solution Overview

The solution follows a progressive agent architecture.

User Screenshot
↓
Level 1 – Audit
↓
Level 2 – Compare
↓
Level 3 – Release Validation
↓
Final Recommendation

Each level increases the degree of analysis and decision support while preserving explainability and human control.

4.System Architecture

The application consists of four primary layers:

Frontend Layer

React-based dashboard providing:

• Screenshot uploads

• Interactive reports

• Findings visualization

• Decision dashboards

Backend Layer

FastAPI services responsible for:

• Image processing

• Classification

• Scoring

• Report generation

AI Layer

Gemini 2.5 Flash generates:

• Executive summaries

• Stakeholder summaries

• Trade-off explanations

• Human-readable recommendations

Governance Layer

Responsible AI safeguards:

• Confidence scoring

• Human review checkpoints

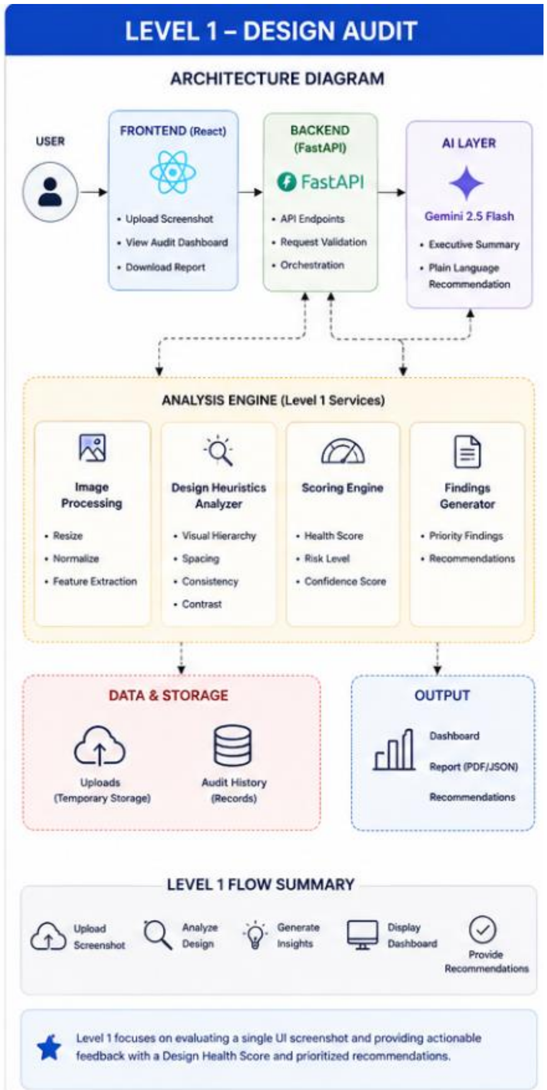
• Validation notes

• Known limitations

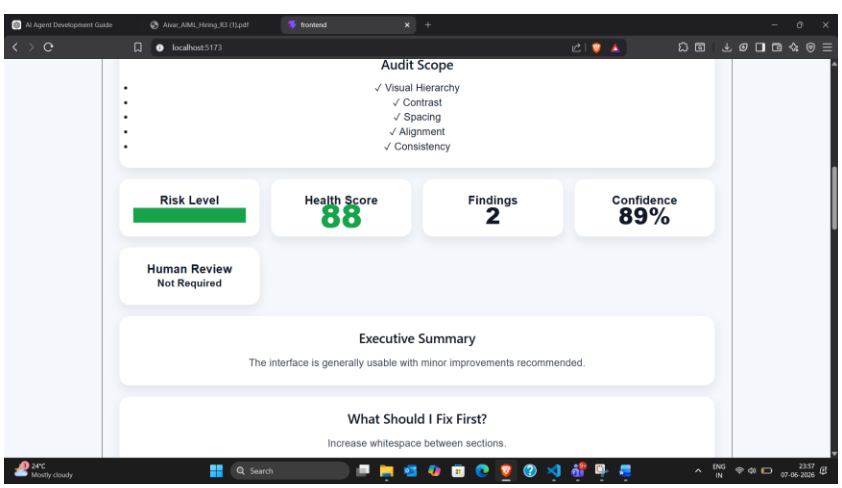
Level 1 – Design Audit Agent

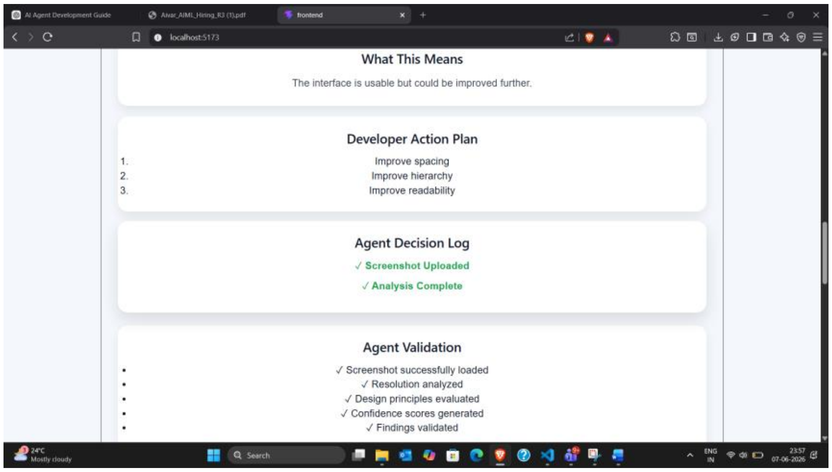
Objective

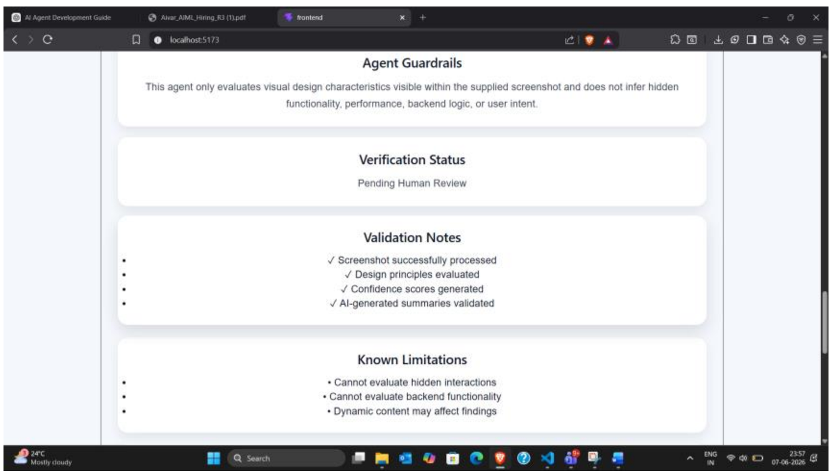
Evaluate a single screenshot and provide actionable UX insights.



SCREENSHOTS:







Inputs

• Screenshot

Outputs

• Design Health Score

• Risk Level

• Executive Summary

• Priority Findings

• Plain Language Recommendation

• Human Review Recommendation

Key Innovation

Rather than simply identifying issues, the system explains their impact on usability and provides recommendations that can be understood by both designers and non-designers.

Example Insights

• Weak visual hierarchy

• Dense layouts

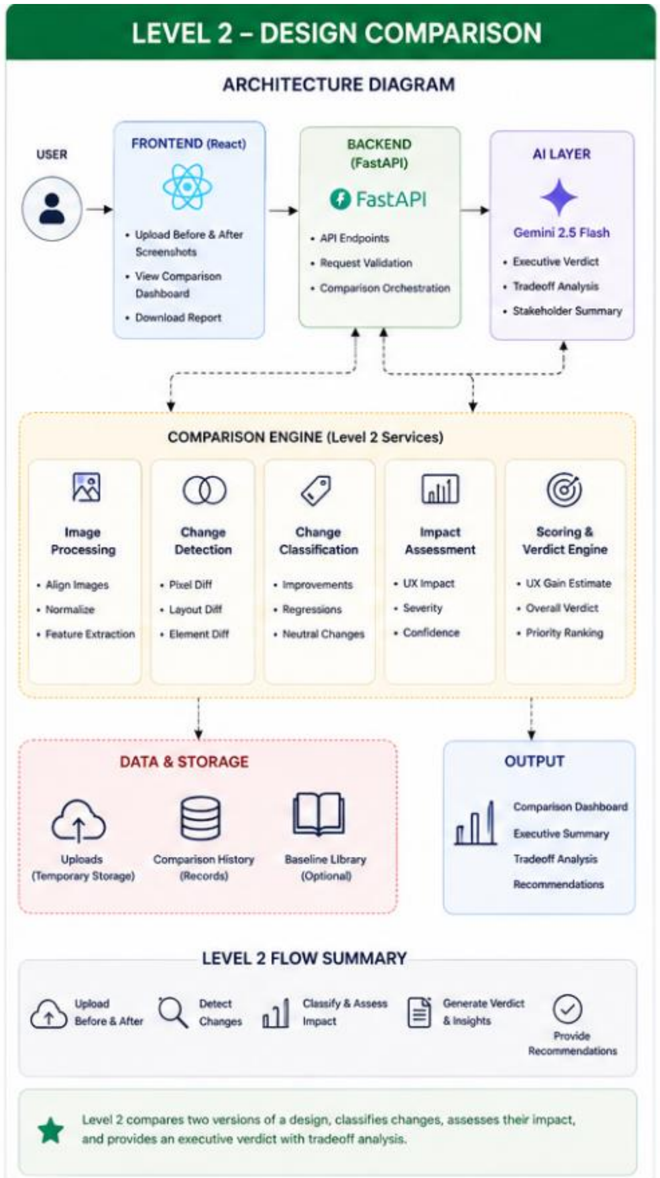
• Low contrast areas

• Consistency concerns

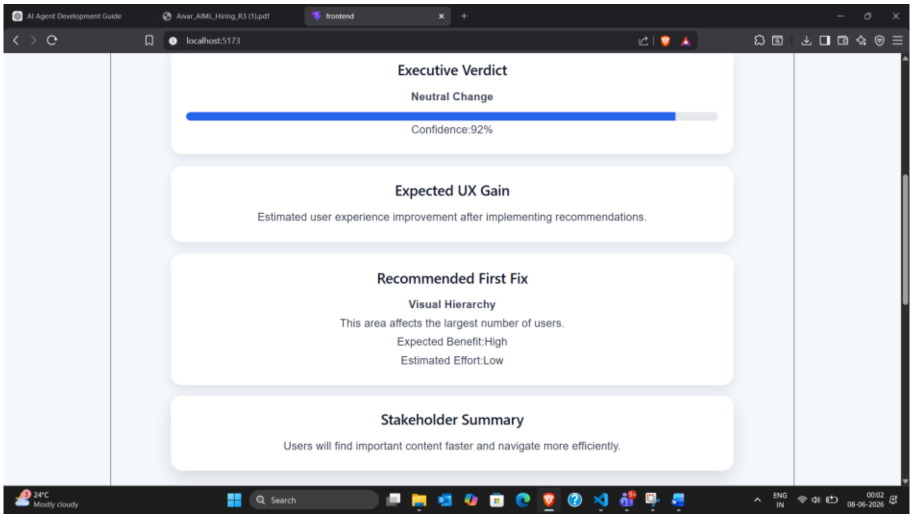
Level 2 – Design Comparison Agent

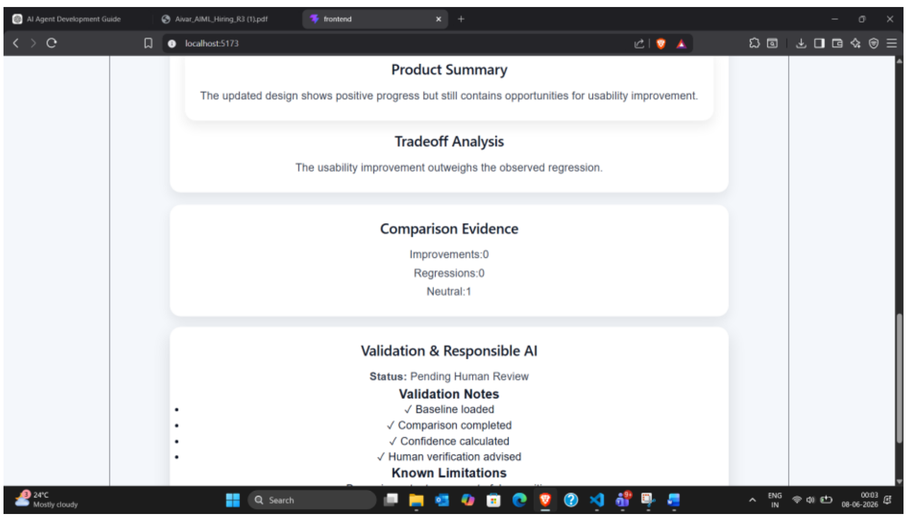
Objective

Determine whether a redesign improves user experience.



SCREENSHOTS:





Inputs

- Before Screenshot
- After Screenshot

Outputs

- Executive Verdict
- UX Gain Estimate
- Priority Fix Matrix
- Stakeholder Summary

• Tradeoff Analysis

Key Feature

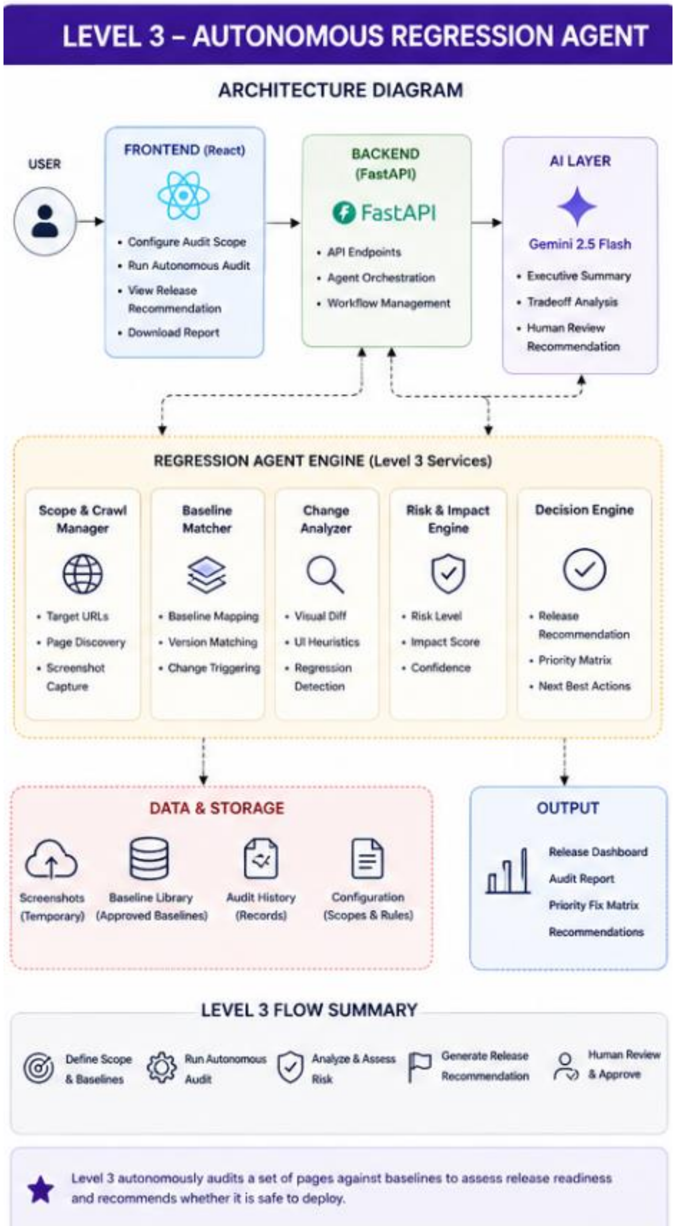
Most tools detect differences.

Design Audit Agent explains whether those differences are beneficial, harmful, or neutral from a usability perspective.

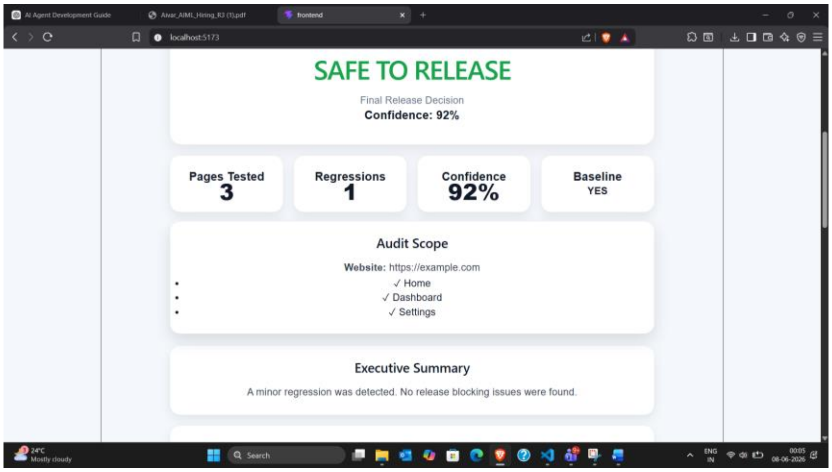
Level 3 – Autonomous Regression Agent

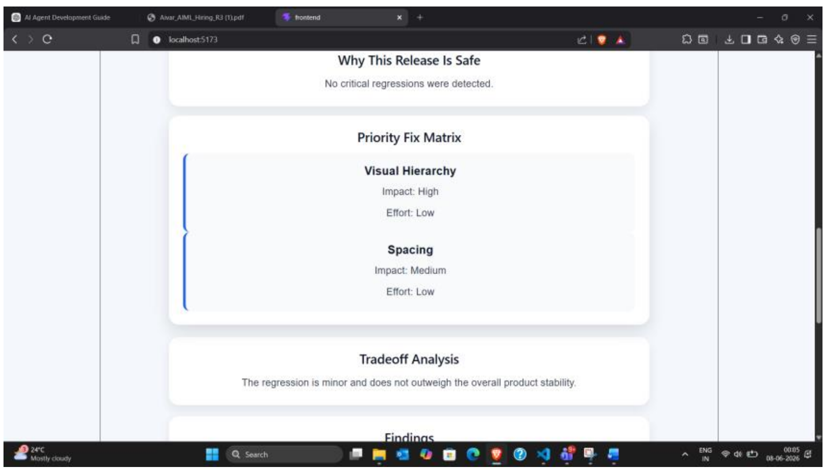
Objective

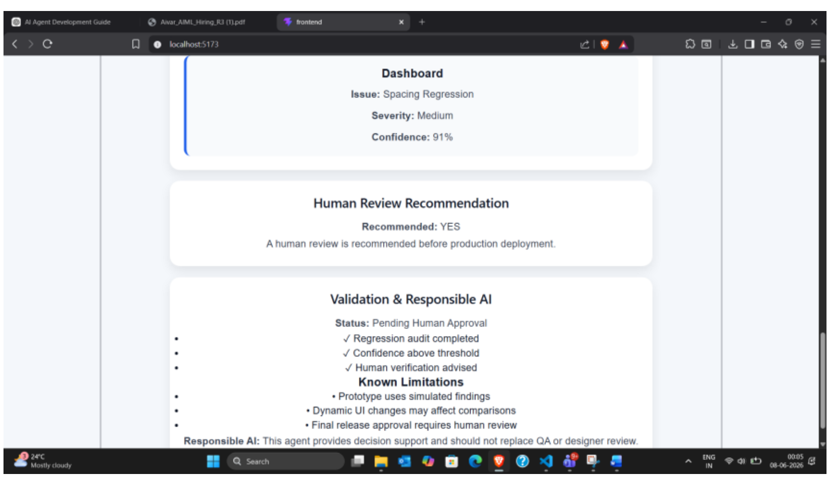
Support release readiness decisions.



SCREENSHOTS:







Inputs

- Candidate Release
- Approved Baselines

Outputs

• Release Recommendation

• Priority Fix Matrix

• Tradeoff Analysis

• Human Review Recommendation

• Responsible AI Validation

Key Feature

This level simulates how an AI agent could participate in release review workflows while maintaining human oversight.

5.Key Design Decisions

Decision	Reason
Use deterministic scoring	Reduce hallucinations
Use Gemini for summaries only	Improve explainability
Human review checkpoints	Support responsible AI
Modular service architecture	Improve maintainability
Progressive Level 1→3 workflow	Demonstrate increasing autonomy

6.Trade-Off Analysis

Chosen Approach	Alternative	Trade-Off
Screenshot-based analysis	Full browser automation	Faster implementation but less coverage
Deterministic findings	Fully AI-generated findings	More reliable but less adaptive
Human approval recommendations	Fully autonomous decisions	Safer but less automated

Gemini explanations	Rule-based summaries	More natural but API dependent

The project intentionally prioritizes explainability and trust over complete automation.

Clear reasoning and human oversight were considered more valuable than maximizing AI autonomy.

7.Edge Cases Handled

The system includes handling for:

• Missing screenshot uploads

• Invalid image formats

• No visual changes detected

• Low-confidence findings

• Dynamic content differences

• Human review requirements

These cases ensure the system remains robust even when ideal inputs are unavailable.

8.Future Enhancements

The following enhancements were identified during development:

• Playwright website crawling

• Automated screenshot capture

• Accessibility scoring

• WCAG compliance validation

• CI/CD integration

• Continuous regression monitoring

• Multi-page autonomous website auditing

Conclusion

Design Audit Agent demonstrates how AI can support UX evaluation workflows through explainable analysis, structured decision-making, and responsible AI practices.

Rather than replacing designers, the system helps teams make better design decisions faster by transforming screenshots into actionable design intelligence.

The guiding principle behind this project is simple:

"Not to replace human judgment, but to augment it through transparent and responsible AI."